

Research Interests

- **Multi-modal Representation Learning:** unified modeling of language, vision, and audio in large-scale foundation models.
- **Embodied AI Systems:** 3D perception and domain adaptation for real-world embodied agents.

🏛 Education

Korea University

Ph.D. in Artificial Intelligence

GPA: 4.28 / 4.5

Advisor: [Prof. Sangpil Kim](#)

Seoul, South Korea

Sep. 2022 – Aug. 2027 (Expected)

Inha University

B.S. in Computer Science

GPA: 4.0 / 4.5

Advisor: [Prof. Jinman Jung](#)

Incheon, South Korea

Mar. 2016 – Aug. 2022

🧰 Research Experience

University of Michigan

Visiting Researcher

Advisors: [Prof. Honglak Lee](#)

Projects: World Model, Vision-Language Action Model

Ann Arbor, MI, USA

Aug. 2026 – Aug. 2027 (Expected)

Samsung Advanced Institute of Technology (SAIT)

Visiting Researcher, Computer Vision Lab

Advisors: [Ph.D. Sujin Jang](#)

Projects: Autonomous Driving, Vision-Language Navigation

Suwon, South Korea

Jan. 2024 – Feb. 2025

📖 Publications († Equal contribution)

Conference Papers

[C9] Training-Free Global Geometric Association for 4D LiDAR Panoptic Segmentation

Gyeongrok Oh, Youngdong Jang, Jonghyun Choi, Suk-Ju Kang, Guang Lin, Sangpil Kim

Submission, 2026 [\[paper\]](#)

[C8] Motion Cues from Image-based Point Tracking for LiDAR Scene Flow Estimation

Gyeongrok Oh[†], Youngdong Jang[†], Jong Wook Kim, Hyunju Ryu, Hyung-gun Chi, SeungHyeon Kim, Seungryong Kim, Jonghyun Choi, Sangpil Kim

NeurIPS, 2026 [\[paper\]](#)

[C7] Active Test-time Vision-Language Navigation

Heeju Ko, Sungjune Kim, Gyeongrok Oh, Jeongyoon Yoon, Honglak Lee, Sujin Jang, Seungryong Kim, Sangpil Kim

NeurIPS, 2025 [\[paper\]](#)

[C6] Test-time Adaptation for Online Vision-Language Navigation with Feedback-based Reinforcement Learning

Gyeongrok Oh[†], Sungjune Kim[†], Heeju Ko, Daehyun Ji, Dongwook Lee, Byung-Jun Lee, Sujin Jang*, Sangpil Kim*

ICML, 2025 [\[paper\]](#)

[C5] 3D Occupancy Prediction with Low-Resolution Queries via Prototype-aware View Transformation

Gyeongrok Oh[†], Sungjune Kim[†], Heeju Ko, Hyung-gun Chi, Jinkyu Kim, Dongwook Lee, Daehyun Ji, Sungjoon Choi, Sujin Jang*, Sangpil Kim*

CVPR, 2025 [\[paper\]](#)

[C4] MEVG: Multi-event Video Generation with Text-to-Video Models

Gyeongrok Oh, Jaehwan Jeong, Sieun Kim, Wonmin Byeon, Jinkyu Kim, Sungwoong Kim, Sangpil Kim

ECCV, 2024 [\[paper\]](#)

[C3] **CMDA: Cross-Modal and Domain Adversarial Adaptation for LiDAR-based 3D Object Detection**
Gyusam Chang†, Wonseok Roh†, Sujin Jang, Dongwook Lee, Daehyun Ji, **Gyeongrok Oh**, Jinsun Park, Jinkyu Kim*, Sangpil Kim*
AAAI, 2024 [paper]

[C2] **Functional Hand Type Prior for 3D Hand Pose Estimation and Action Recognition from Egocentric View Monocular Videos**
Wonseok Roh, Seung hyun Lee, Won Jeong Ryoo, **Gyeongrok Oh**, Jakyung Lee, Soo Yeon Hwang, Hyung-gun Chi, Sangpil Kim
BMVC (Oral), 2023 [paper]

[C1] **Sound-Guided Semantic Video Generation**
Seung hyun Lee, **Gyeongrok Oh**, Wonmin Byeon, Chanyoung Kim, Won Jeong Ryoo, Sang Ho Yoon, Jihyun Bae, Jinkyu Kim*, Sangpil Kim*
ECCV, 2022 [paper]

Journal Papers

[J4] **LVMark: Robust Watermark for Latent Video Diffusion Models**
Youngdong Jang*, MinHyuk Jang*, JaeHyeok Lee, Feng Yang, **Gyeongrok Oh**, Jongheon Jeong, Sangpil Kim
IEEE Transactions on Information Forensics & Security, 2026. IF: 8.0 [paper]

[J3] **FPANet: Frequency-based Video Demoireing using Frame-level Post Alignment**
Gyeongrok Oh, Sungjune Kim, Heon Gu, Sang Ho Yoon, Jinkyu Kim, Sangpil Kim
Neural Networks, 2025. IF: 6.3 [paper]

[J2] **Robust Sound-Guided Image Manipulation**
Seunghyun Lee†, Hyung-gun Chi†, **Gyeongrok Oh**, Wonmin Byeon, Sang Ho Yoon, Hyunje Park, Wonjun Cho, Jinkyu Kim*, Sangpil Kim*
Neural Networks, 2024. IF: 6.3 [paper]

[J1] **Audio-Guided Implicit Neural Representation for Local Image Stylization**
Seung Hyun Lee†, Sieun Kim†, Wonmin Byeon, **Gyeongrok Oh**, Sumin In, Hyeongcheol Park, Sang Ho Yoon, Sung-hee Hong, Jinkyu Kim*, Sangpil Kim*
Computational Visual Media, 2024. IF: 18.3 [paper]

💡 Patents

[P1] **Method and Apparatus with 3D Occupancy Prediction Learning** US Patent
Gyeongrok Oh, Sujin Jang, Sangpil Kim, Sungjune Kim, Jinkyu Kim, Dongwook Lee, Dae Hyun Ji
US20250356594A1, 2025 (Pending)

👤 Teaching Experience

Korea University Seoul, South Korea
Teaching Assistant

- Statistics (2023), Computer Vision (2025), Intelligent Mobility (2025)

LG CNS Seoul, South Korea
Teaching Assistant

- Anomaly Detection Course for Executives and Employees (2022, 2023)

Inha University Incheon, South Korea
Teaching Assistant

- Data Structures (2021)

🏆 Awards & Honors

BK21 Academic Excellence Scholarship Korea University

Departmental Scholarship Inha University
Department of Computer Science 2020–2022

Reviewer: CVPR (2024–2026), ICCV (2025), ECCV (2026), NeurIPS (2026), WACV (2023, 2025–2026), BMVC (2026)
Program Committee: AAAI (2025–2026)